Criticality and phase diagram study of the long-range quantum Ising chain

Mini-Meeting

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Introduction

Introduction

A variety of quantum models can be simulated

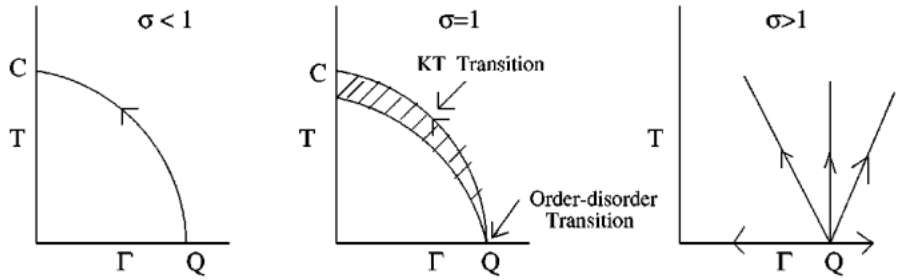
- Using Rydberg atoms, dipolar quantum gases, polar molecules, multimode cavities, trapped ions ...
- There is great interest in equilibrium and dynamical properties of long-range models
  - Ising and/or XY quantum spin chains with tunable LR interactions
    - Be ions stored in a Penning trap
    - Neutral atoms coupled to photonic modes of a cavity
    - Trapped ions coupled to motional degrees of freedom

Introduction

There is great interest in equilibrium and dynamical properties of long-range models

- The critical behavior for a quantum spin chain with $J_{ij} = \frac{J}{r_{ij}^{1+\sigma}}$

$$\hat{H} = -\sum_{ij} J_{ij}\sigma_i^z\sigma_j^z - \Gamma\sum_i \sigma_i^x$$



- Quantum transitions with values of $\sigma \leq 2/3 \rightarrow$ mean-field, $\sigma > 2/3 \rightarrow$ nontrivial exponents.
- The finite-temperature transition is determined by the thermal fixed point

[Dutta and Bhattacharjee, 2001]

Model and Methods

1. The model
2. Quantum to Classical
3. Path Integral Monte Carlo
4. Criticality and finite-size systems

The model

System Hamiltonian

$$\hat{H} = -\frac{V}{K}\sum_{i<j}\frac{\hat{S}_i^z\hat{S}_j^z}{r_{ij}^{\alpha}} - h\sum_i \hat{S}_i^x.$$

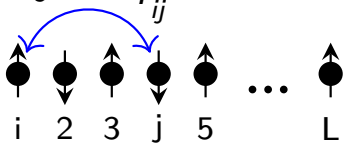
$$K = \sum_{i<j} r_{ij}^{-\alpha}$$

$$\hat{S}_i^x = \frac{\hbar}{2}\sigma_i^x$$

Reduced units Hamiltonian

$$\hat{H} = -\sum_{i<j} V\frac{\hat{S}_i^z\hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

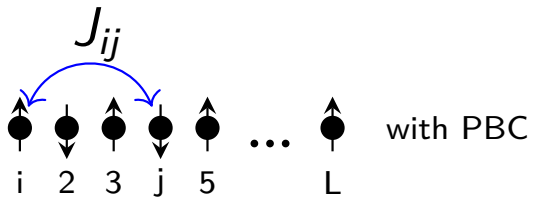
$$J_{ij} = \frac{V}{r_{ij}^{\alpha}}$$



with PBC

Quantum to Classical mapping

$$\hat{H} = \underbrace{\sum_{i<j} J_{ij} \hat{S}_i^z \hat{S}_j^z}_{\hat{H}_0} + \underbrace{\sum_i \hat{S}_i^x}_{\hat{H}_1}$$



Partition Function

$$Z = \sum_{S_0} \langle S_0 | e^{-\beta \hat{H}} | S_0 \rangle \cong \sum_{S_0, \ldots, S_{M-1}} \langle S_0 | \hat{U}_H | S_1 \rangle \langle S_1 | \hat{U}_H | S_2 \rangle \ldots \langle S_{M-1} | \hat{U}_H | S_0 \rangle$$

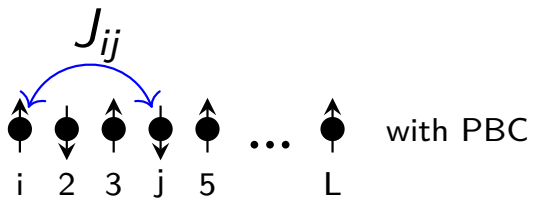
where $\hat{U}_H = e^{-\epsilon \hat{H}_0} e^{-\epsilon \hat{H}_1}$ with $\epsilon = \beta / M$

Matrix element

$$\langle S_k | e^{-\epsilon \hat{H}_0} e^{-\epsilon \hat{H}_1} | S_{k+1} \rangle = e^{-\epsilon \sum_{ij} J_{ij} S_{ki}^z S_{kj}^z} \langle S_k | e^{-\epsilon \hat{H}_1} | S_{k+1} \rangle$$

Quantum to Classical mapping

$$\hat{H} = \underbrace{\sum_{i<j} J_{ij} \hat{S}_i^z \hat{S}_j^z}_{\hat{H}_0} + \underbrace{\sum_i \hat{S}_i^x}_{\hat{H}_1}$$



Matrix element

$$\langle S_k | e^{-\epsilon \hat{H}_0} e^{-\epsilon \hat{H}_1} | S_{k+1} \rangle = e^{-\epsilon \sum_{ij} J_{ij} S_{ki}^z S_{kj}^z} \langle S_k | e^{-\epsilon \hat{H}_1} | S_{k+1} \rangle$$

Looking for a classical partition function

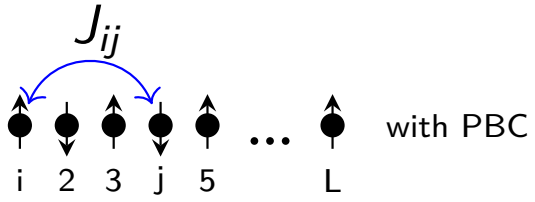
$$e^{-\epsilon \hat{H}_1} = \sum_i (\cosh(\epsilon) - \hat{S}_i^x \sinh(\epsilon)) \qquad \langle S_{ki}^z | e^{-\epsilon \hat{S}_i^x} | S_{(k+1)i}^z \rangle \equiv \Lambda e^{-\gamma S_{ki}^z S_{(k+1)i}^z}$$

$$\langle \uparrow (\downarrow)_i | e^{-\epsilon \hat{S}_i^x} | \uparrow (\downarrow)_i \rangle = \cosh(\epsilon) \equiv \Lambda e^{-\gamma}$$

$$\langle \uparrow (\downarrow)_i | e^{-\epsilon \hat{S}_i^x} | \downarrow (\uparrow)_i \rangle = \sinh(\epsilon) \equiv \Lambda e^{+\gamma}$$

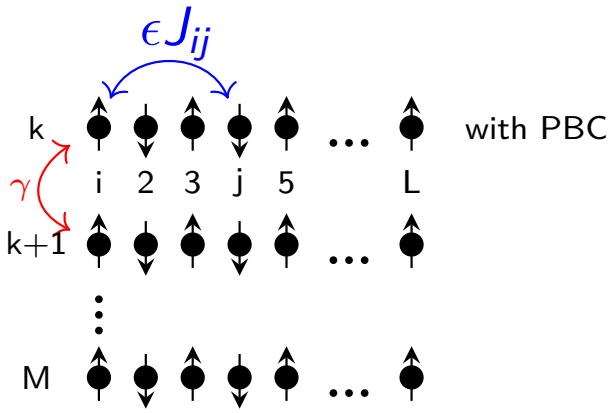
Quantum to Classical mapping

$$\hat{H} = \underbrace{\sum_{i<j} J_{ij} \hat{S}_i^z \hat{S}_j^z}_{\hat{H}_0} + \underbrace{\sum_i \hat{S}_i^x}_{\hat{H}_1}$$



Classical, anisotropic, 2D Ising model

$$Z = \left(\prod_{i=1}^{L} \Lambda\right)^M \sum_{S_{ki}^z} exp\left[-\sum_{k=1}^{M}\sum_{i,j=1}^{L} \epsilon J_{ij} S_{ki}^z S_{kj}^z - \sum_{i=1}^{M} \gamma \sum_{k=1}^{L} S_{ki}^z S_{(k+1)i}\right]$$



Path Integral Monte Carlo

PIMC

Numerically exact technique for the study of unfrustrated systems of bosons and quantum spins.

- Exploits the quantum to classical mapping
- Properties of the quantum system are accessed via conventional Metropolis-type Monte Carlo
- The accuracy of the mapping is exact up to $O((\beta/M)^2)$ corrections,which vanish in the limit $M \to \infty$

Our implementation

Combine:

- Wolff cluster updates [Wolff, 1989] in imaginary time
- Long-range cluster updates [Luijten and Blöte, 1997] in the spatial direction

Criticality and finite-size systems

PIMC gives direct access to observables like

Binder cumulant

$$U = \frac{1}{2}\left[3 - \frac{\langle m_z^4 \rangle}{\langle m_z^2 \rangle^2}\right]$$

Susceptibility

$$\chi = \beta L(\langle m_z^2 \rangle - \langle |m_z| \rangle^2)$$

The finite-size scaling hypothesis

$$|t| \sim \xi^{-1/\nu}$$
$$Q \sim \xi^{\kappa/\nu}$$

$$|t_{max}(L)| \sim L^{-1/\nu}$$
$$Q_{max}(L) \sim L^{\kappa/\nu}$$

The hypothesis

$$Q(t, L) = L^{\sigma} f(\xi/L)$$

Results

1. Ground state phase transition
2. Finite temperature phase transition
3. Criticality and phase diagram

Ground state phase transition

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

Binder cumulant

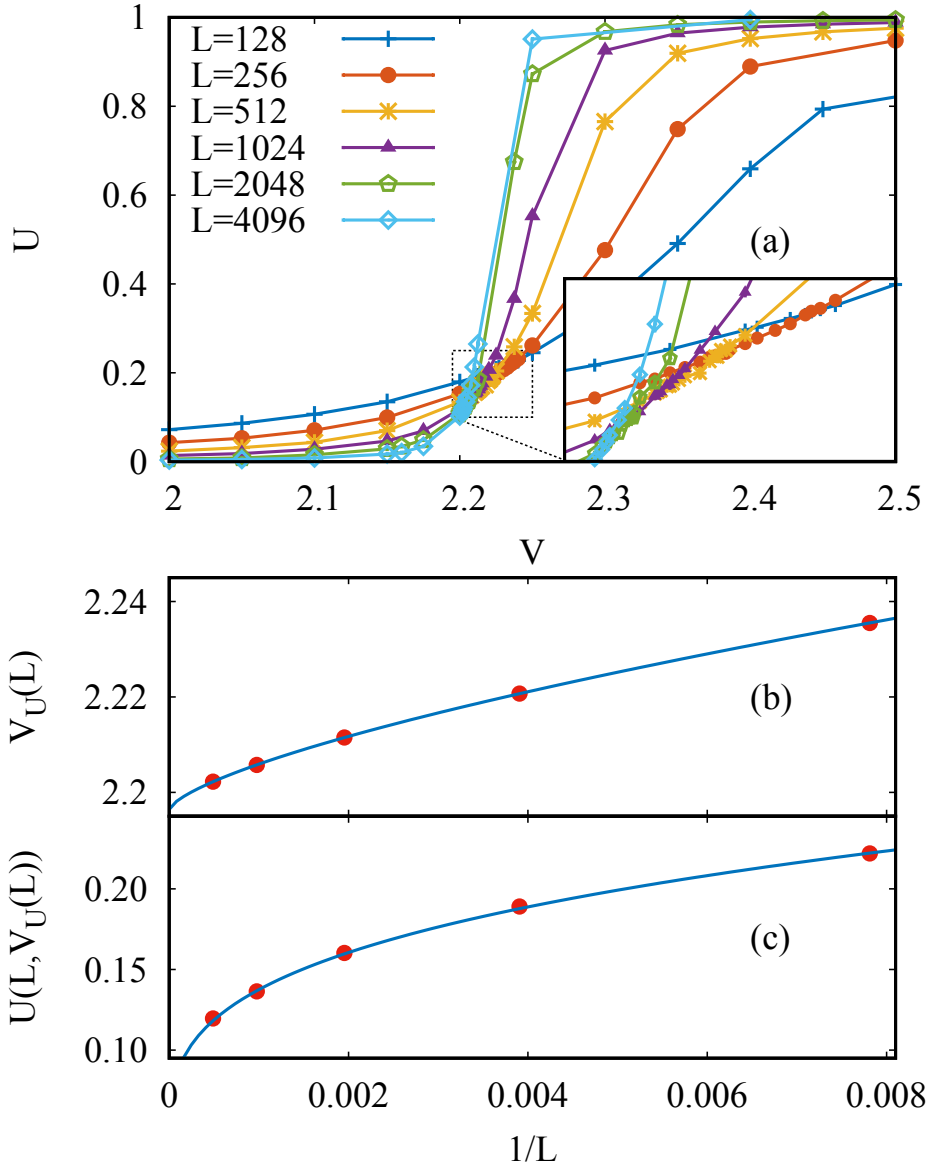
$$U = \frac{1}{2}\left[3 - \frac{\langle m_z^4 \rangle}{\langle m_z^2 \rangle^2}\right]$$

FSS

$$g_U(L) = g_c\left(1 + aL^{-\omega-\frac{1}{\nu}}\right)$$

$$U(L, g_U(L)) = b + cL^{-\omega}$$

[Angelini et al., 2014]



Finite temperature phase transition

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

Binder cumulant

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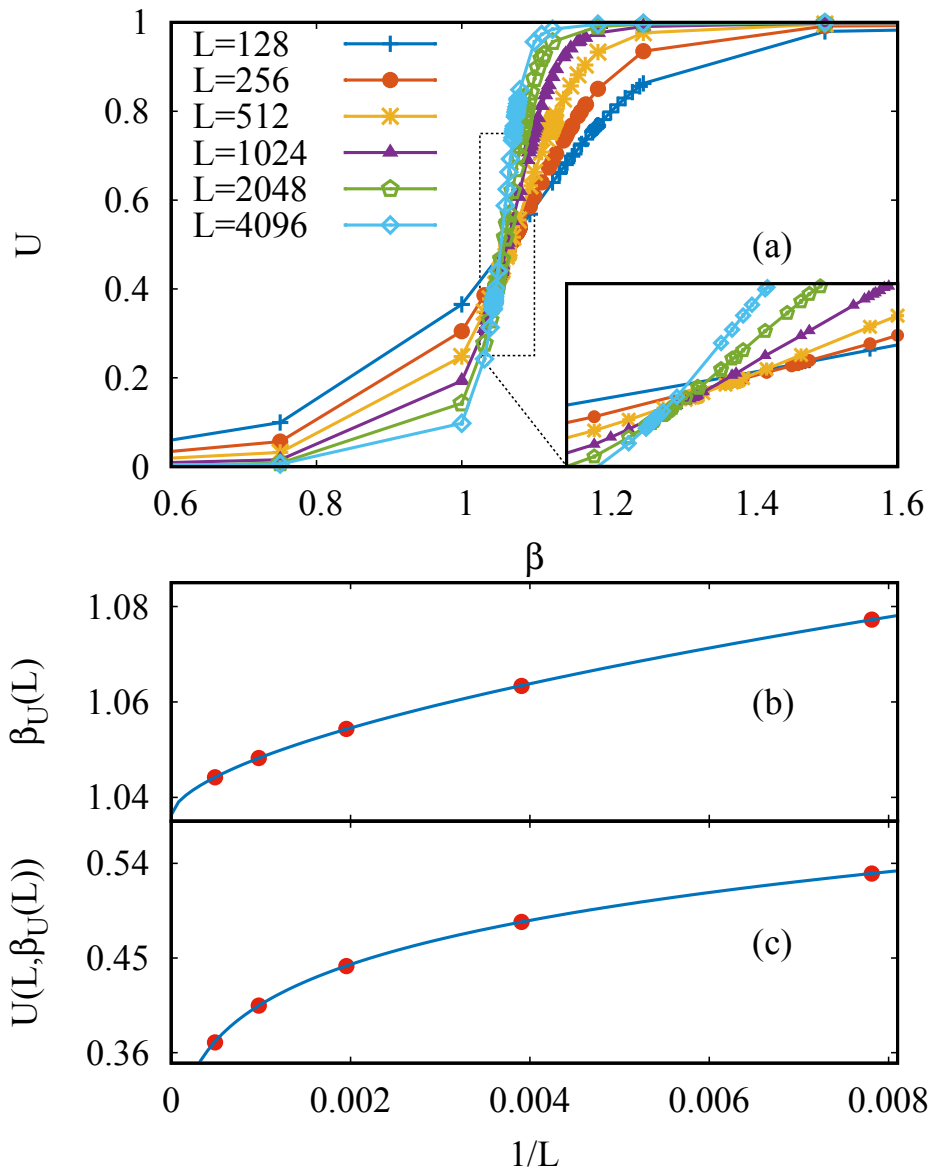


Finite temperature phase transition

Hamiltonian

$$\hat{H} = -\sum_{i<j} V \frac{\hat{S}_i^z \hat{S}_j^z}{r_{ij}^{\alpha}} - \sum_i \hat{S}_i^x$$

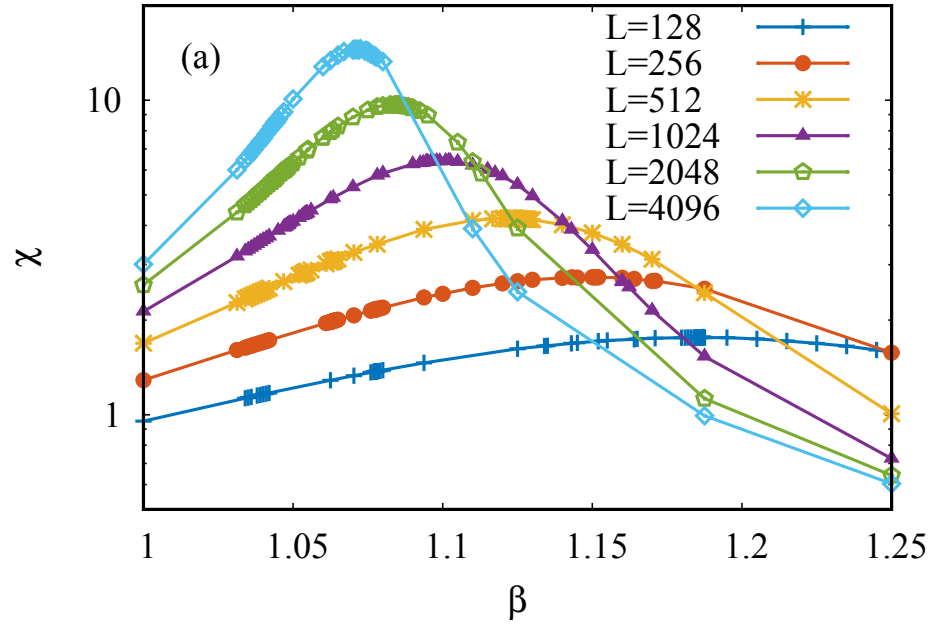
Susceptibility

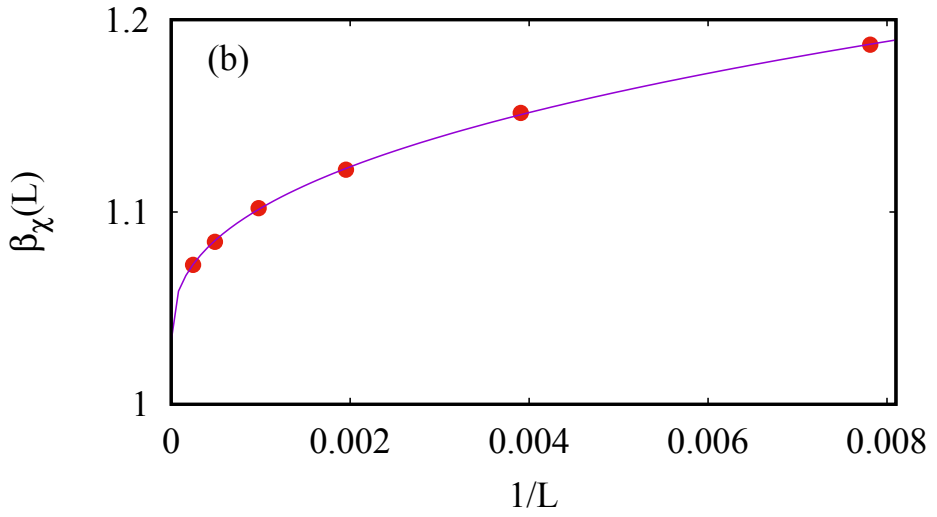
$$\chi = \beta L(\langle m_z^2 \rangle - \langle |m_z| \rangle^2)$$

FSS

$$g_\chi(L) = g_c + fL^{-1/\nu}$$

[Sandvik, 2010]





Finite temperature phase transition

Curve collapse analysis

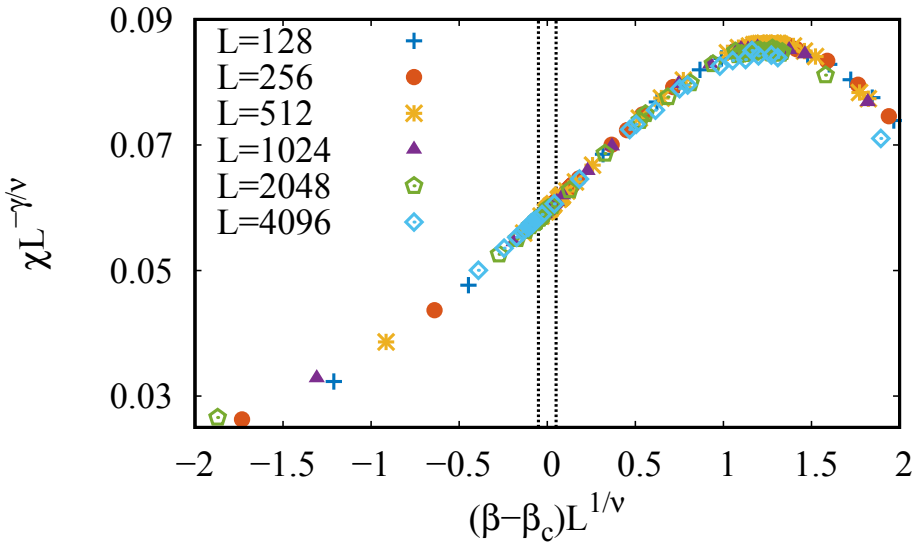
$$\chi \sim L^{\frac{\gamma}{\nu}} \cdot f(L^{-\frac{1}{\nu}}(g - g_c))$$

Perform a high order polynomial fit

$$y_l \equiv \chi(L) L^{-\gamma/\nu}$$

$$x_l \equiv (g - g_c) L^{1/\nu}$$

[Sandvik, 2010]



Criticality and phase diagram

Hamiltonian

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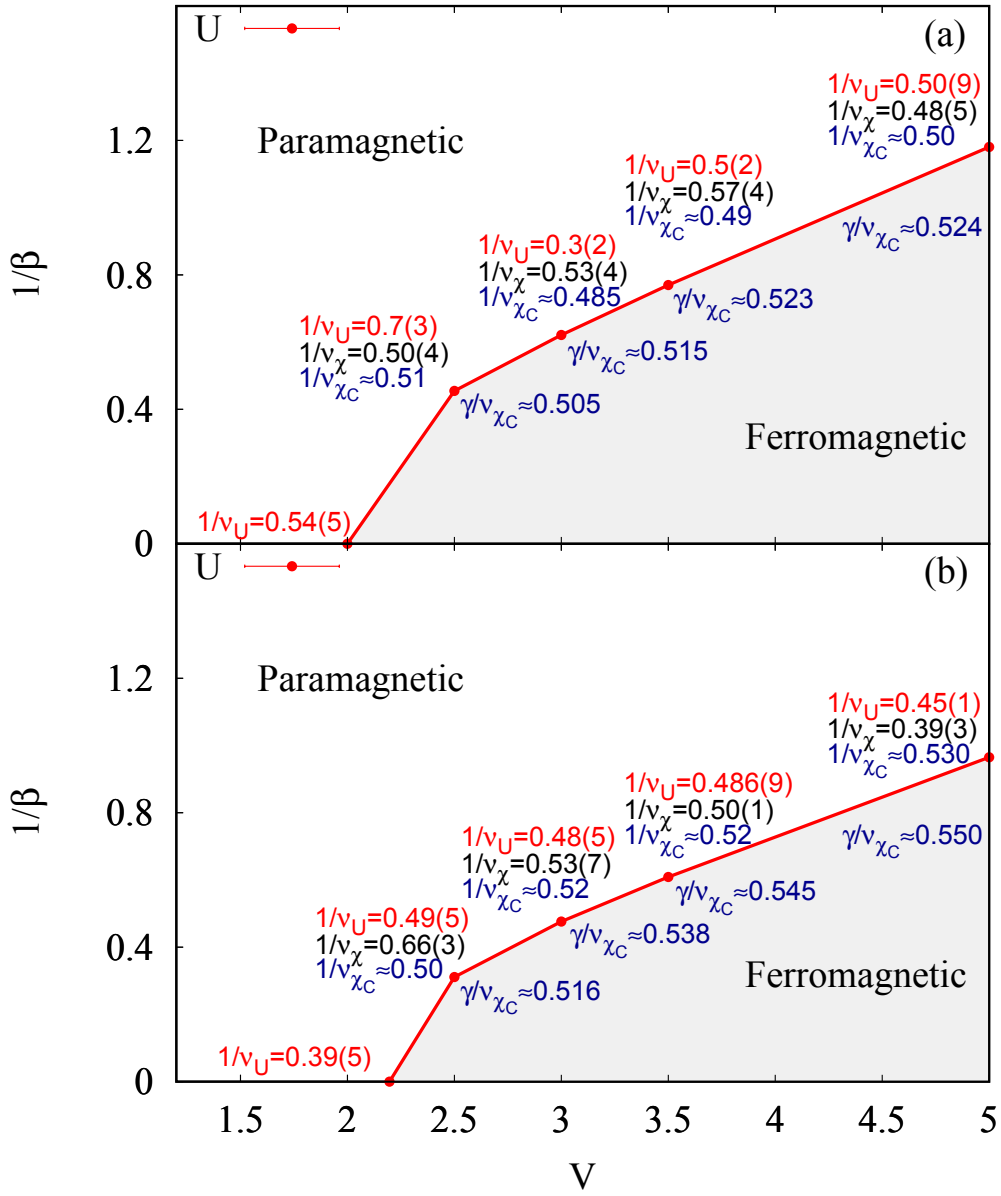

Conclusions

- Studied the quantum Ising chain with algebraic decaying interaction in a regime of parameters
- No previous results for $\alpha < 1$ were present in the literature
- With these **preliminary** results:
  - The finite$-$T transition probably is in the same universality class for all values of α, not the case for the ground state transition

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Thank you for your attention

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Bibliography

- Angelini, M. C., Parisi, G., and Ricci-Tersenghi, F. (2014). Relations between short-range and long-range Ising models. *Physical Review E*, 89(6):062120.
- Dutta, A. and Bhattacharjee, J. K. (2001). Phase transitions in the quantum ising and rotor models with a long-range interaction. *Phys. Rev. B*, 64:184106.
- Luijten, E. and Blöte, H. W. J. (1997). Classical critical behavior of spin models with long-range interactions. *Phys. Rev. B*, 56:8945–8958.
- Sandvik, A. W. (2010). Computational studies of quantum spin systems. *AIP Conference Proceedings*, 1297(1):135–338.
- Wolff, U. (1989). Collective monte carlo updating for spin systems. *Phys. Rev. Lett.*, 62:361–364.